

Dynamic Programming & Reinforcement Learning

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MDP Recap and Notation



Recap: the ingredients of an MDP

For MDP value calculations, we use $(\mathcal{X}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$.

$\mathcal{X}, \mathcal{A}$	Sets of states and available actions
$\mathcal{P}(\cdot \mid x, a)$	Distribution of the next state after (x, a)
$\mathcal{R}(\cdot \mid x, a)$	Distribution of the reward after (x, a)
γ	Discount factor: how much future rewards count

A policy π selects actions; ρ is the initial-state distribution.

$$X_1 \sim \rho, \quad \underbrace{X_t}_{\text{observe}} \longrightarrow \underbrace{A_t}_{\text{choose}} \longrightarrow \underbrace{(R_t, X_{t+1})}_{\text{receive reward and next state}} .$$

The joint law of (R_t, X_{t+1}) has marginals $\mathcal{R}$ and $\mathcal{P}$; reward and next state need not be independent. We use finite nonempty state/action sets and bounded rewards. Continuing tasks use $0 \leq \gamma < 1$; finite-horizon and episodic cases are identified separately.

Reading the symbols

X_t, A_t, R_t

Random state, action, and reward at time t

x, a, x'

Particular state, action, and next-state values; x' is not a derivative

$x \in \mathcal{X}, f : \mathcal{X} \rightarrow \mathbb{R}$

x belongs to $\mathcal{X}$; f assigns a real number to each state

$X_1 \sim \rho$

X_1 is drawn from the distribution ρ

$| x, \mathcal{P}(\cdot | x, a)$

“Given x ”; the dot leaves the outcome unspecified, indicating the whole distribution

$\triangleq, \mathbb{E}[\cdot]$

“Defined as”; expectation, a probability-weighted average

In a quantity such as r^π , the superscript labels the policy; it is not a power. For a fixed model input (x, a) , draw $R \sim \mathcal{R}(\cdot | x, a)$ and define $r(x, a) \triangleq \mathbb{E}[R]$. The reward R is random; $r(x, a)$ is its mean, even at unvisited inputs.

Markov models and policies

The *Markov property*: given the current state and action, the joint law of next state and reward ignores earlier history, including past rewards. In particular,

$$\Pr(X_{t+1} = x' \mid X_1 = x_1, A_1 = a_1, \dots, X_t = x_t, A_t = a_t) = \mathcal{P}(x' \mid x_t, a_t).$$

Stationary dynamics means this joint law does not change with t .

A stationary Markov policy uses the same rule at every time: even given the whole observed history, its action probabilities are $\pi(a \mid X_t)$.

$$\pi(a \mid x) = \Pr(A_t = a \mid X_t = x), \quad \sum_{a \in \mathcal{A}} \pi(a \mid x) = 1.$$

The displayed conditionals require positive-probability conditioning events; the model and policy kernels are specified at every valid input.

For a deterministic policy, $\pi(x)$ is the chosen action: $\pi(a \mid x) = 1$ if $a = \pi(x)$, and 0 otherwise. A stochastic policy can randomize.

A Markov environment can still be controlled by a history-dependent policy. Here, π is stationary Markov unless stated otherwise.

A kernel assigns a distribution to each input

A distribution assigns probabilities to sets of possible values. Here $\Pr$ denotes probability; $\mathcal{P}$ names the transition model. A *probability kernel* gives a distribution for each conditioning input:

$$(x, a) \longmapsto \mathcal{P}(\cdot \mid x, a).$$

For example, with two possible next states u, v :

Input	Next state u	Next state v
(u, wait)	0.8	0.2
(u, move)	0.1	0.9

Each row is one distribution: nonnegative entries summing to one. The kernel specifies all such rows. A policy is also a kernel, from states to actions.

A deterministic transition $f(x, a)$ returns one state; its kernel puts probability one there.

“Transition function” may also mean a kernel, depending on the text.

An expectation is a probability-weighted average

Fix x , draw $A \sim \pi(\cdot | x)$, then $R \sim \mathcal{R}(\cdot | x, A)$. Suppose the action probabilities and mean rewards are

a	a_1	a_2
$\pi(a x)$	0.25	0.75
$r(x, a)$	2	6

The mean reward under the policy is

$$\begin{aligned} r^\pi(x) &\triangleq \sum_{a \in \mathcal{A}} \pi(a | x) r(x, a) = \mathbb{E}[R] \\ &= 0.25(2) + 0.75(6) = 5. \end{aligned}$$

First average rewards for each action, then average over the policy's actions. This is the law of total expectation: average the conditional averages.

What does $\pi(da \mid x)$ mean?

Fix x , draw $A \sim \pi(\cdot \mid x)$, and take bounded measurable $h : \mathcal{A} \rightarrow \mathbb{R}$:

$$\mathbb{E}[h(A)] = \int_{\mathcal{A}} \pi(da \mid x) h(a).$$

$\pi(da \mid x)$ says which distribution supplies the weights and which variable is averaged out. The d is part of integration notation, not a derivative of π .

Finite actions: use the probability of each action.

$$\int_{\mathcal{A}} \pi(da \mid x) h(a) = \sum_{a \in \mathcal{A}} \pi(a \mid x) h(a).$$

Continuous actions with a density p_π : use an ordinary integral.

$$\int_{\mathcal{A}} \pi(da \mid x) h(a) = \int_{\mathcal{A}} p_\pi(a \mid x) h(a) da.$$

Measurability is automatic in the finite case. A density gives probabilities by integration, not by its value at a point. The kernel notation also covers discrete and mixed distributions.

Two-stage averaging: action, then next state

Fix x , draw $A \sim \pi(\cdot | x)$, then $X' \sim \mathcal{P}(\cdot | x, A)$. For bounded $h : \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}[h(X')] = \sum_a \pi(a | x) \underbrace{\sum_{x'} \mathcal{P}(x' | x, a) h(x')}_{\text{mean for a fixed action}}.$$

This is the tower property: average the conditional means. For the earlier transition table at u , let both actions have probability $1/2$, and let $h(u) = 10$, $h(v) = 0$:

$$\frac{1}{2} \underbrace{[0.8(10) + 0.2(0)]}_8 + \frac{1}{2} \underbrace{[0.1(10) + 0.9(0)]}_1 = 4.5.$$

The general notation for the same nested average is

$$\int_{\mathcal{A}} \pi(da | x) \int_{\mathcal{X}} \mathcal{P}(dx' | x, a) h(x').$$

The joint weight $\pi(a | x) \mathcal{P}(x' | x, a)$ uses the probability chain rule, not independence of the action and next state.

Returns and Value Functions



Finite-horizon returns

Suppose the interaction lasts T decisions. For rewards $(R_1, \dots, R_T)$, the undiscounted return is

$$G^\pi \triangleq R_1 + R_2 + \dots + R_T.$$

With a discount factor $0 \leq \gamma \leq 1$,

$$G^\pi \triangleq R_1 + \gamma R_2 + \dots + \gamma^{T-1} R_T = \sum_{t=1}^T \gamma^{t-1} R_t.$$

The superscript records which policy generates the reward sequence. Even for a fixed policy, different trajectories can have different returns.

The discount factor

$$G^\pi = R_1 + \gamma R_2 + \cdots + \gamma^{T-1} R_T.$$

- A reward $t - 1$ steps away receives weight γ^{t-1} .
- At $\gamma = 0$, only the first reward contributes.
- At $\gamma = 1$, all T rewards have equal weight.
- A smaller γ gives less weight to future rewards.

γ specifies the task objective; it is not an algorithmic step size.

Check. If the horizon is infinite, what can happen when every reward is 1 and $\gamma = 1$?

Expected return defines a value

Write $\mathbb{E}_x^\pi$ for expectation when we initialize $X_1 = x$ and follow π . This experiment is defined even if an earlier initial law has $\rho(x) = 0$.

$$V^\pi(x) \triangleq \mathbb{E}_x^\pi \left[\sum_{t=1}^T \gamma^{t-1} R_t \right].$$

G^π : random return, $V^\pi(x)$: its mean in this experiment.

For a finite-horizon Markov policy, $\mathbb{E}_{t,x}^\pi$ starts at decision t in state x , using the remaining rules $\pi_t, \dots, \pi_T$:

$$V_t^\pi(x) = \mathbb{E}_{t,x}^\pi \left[\sum_{k=t}^T \gamma^{k-t} R_k \right], \quad V_{T+1}^\pi(x) = 0.$$

Discounting starts from the current decision. The same state can have a different value when fewer decisions remain. The initial value above is V_1^π .

One decision: value and expected reward

When $T = 1$, $V^\pi(x) = \mathbb{E}_x^\pi[R_1]$. Recall that $r(x, a)$ is the mean reward generated at (x, a) . Average over the first action:

$$r^\pi(x) \triangleq \int_{\mathcal{A}} \pi(da \mid x) r(x, a) = \sum_{a \in \mathcal{A}} \pi(a \mid x) r(x, a).$$

The integral and sum express the same probability-weighted average. Thus

$$V^\pi(x) = r^\pi(x) \leq \max_{a \in \mathcal{A}} r(x, a).$$

Choose an action with the largest mean reward with probability one to attain this bound. Randomization cannot improve on it.

For a longer horizon, value also includes future rewards: maximizing the immediate reward alone need not be optimal.

Episodes and termination

An episode ends on reaching a terminal state. The number of decisions T may be random. Here R_T is the final reward and X_{T+1} is terminal.

$$G^\pi = \sum_{t=1}^T \gamma^{t-1} R_t, \quad V^\pi(x) = \mathbb{E}_x^\pi[G^\pi].$$

After termination, a dummy action keeps the system in the terminal state and yields reward zero. This is an *absorbing* state, with value zero. The infinite sequence therefore includes R_T only once, followed by zeros.

For bounded rewards:

- If $\gamma < 1$, discounted returns are bounded.
- If $\gamma = 1$, finite expected episode length is a sufficient condition for finite expected return.

Continuing discounted tasks

For interaction without a final decision time, define

$$G_t^\pi \triangleq \sum_{k \geq t} \gamma^{k-t} R_k, \quad 0 \leq \gamma < 1.$$

The task may model a continuing process or approximate a very long finite horizon.

If $|R_k| \leq R_{\max}$, then

$$|G_t^\pi| \leq R_{\max} \sum_{j=0}^{\infty} \gamma^j = \frac{R_{\max}}{1-\gamma}.$$

From here, use stationary continuing tasks: there is no fixed deadline. Under a stationary Markov policy, the future from x has the same law at every time, so the continuing value needs no decision-time index.

State values and action values

For the continuing task, $\mathbb{E}_x^\pi$ initializes at x and follows π . Write $\mathbb{E}_{x,a}^\pi$ for a different experiment: initialize at x , force the first action to a , then follow π from decision 2.

$$V^\pi(x) \triangleq \mathbb{E}_x^\pi \left[\sum_{t=1}^{\infty} \gamma^{t-1} R_t \right],$$

$$Q^\pi(x, a) \triangleq \mathbb{E}_{x,a}^\pi \left[\sum_{t=1}^{\infty} \gamma^{t-1} R_t \right].$$

- For V^π , follow π from the first decision.
- For Q^π , fix the first action to a , then follow π .

These definitions cover every model input, even if $\rho(x) = 0$ or $\pi(a | x) = 0$; they do not condition on an impossible event.

Interpreting the two value functions

$$V^\pi : \mathcal{X} \rightarrow \mathbb{R}, \quad Q^\pi : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}.$$

- $V^\pi(x)$ scores the policy from a starting state.
- $Q^\pi(x, a)$ scores a first action with that policy as the continuation.

Average over the first action under π , including deterministic policies as a special case:

$$V^\pi(x) = \int_{\mathcal{A}} \pi(da \mid x) Q^\pi(x, a).$$

At $\gamma = 0$, $Q^\pi(x, a) = r(x, a)$. At $\gamma > 0$, action values also include the consequences of the first action.

Comparing policies

For stationary Markov policies, define

$$\pi \geq \pi' \iff V^\pi(x) \geq V^{\pi'}(x) \text{ for every } x.$$

An optimal policy satisfies

$$\pi^* \geq \pi \text{ for every policy } \pi \text{ in the comparison class.}$$

Different optimal policies can have the same value function. Two arbitrary policies may be incomparable: each can be better at different states.

For any initial distribution ρ , pointwise dominance also implies

$$\mathbb{E}_{X_1 \sim \rho}[V^\pi(X_1)] \geq \mathbb{E}_{X_1 \sim \rho}[V^{\pi'}(X_1)].$$

Optimal values

For Π , the class of stationary Markov policies, write

$$V^*(x) = \sup_{\pi \in \Pi} V^\pi(x), \quad Q^*(x, a) = \sup_{\pi \in \Pi} Q^\pi(x, a).$$

The supremum $\sup$ is the least upper bound. Unlike a maximum, it does not assume that a policy attains the bound. If one policy attains the best value at every state, then

$$V^* = V^{\pi^*}.$$

There are two questions to resolve:

1. Does such a policy exist?
2. Can a history-dependent or time-dependent policy do better?

For finite discounted MDPs, a deterministic stationary optimal policy exists. We will connect this fact to Bellman equations.

Bellman Expectation Equations



The return has a recursive form

$$\begin{aligned} G_t^\pi &= R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots \\ &= R_t + \gamma (R_{t+1} + \gamma R_{t+2} + \dots) \\ &= \boxed{R_t + \gamma G_{t+1}^\pi.} \end{aligned}$$

Both sides are random variables. Repeated runs from the same state have the same return distribution under the same policy, but need not have the same realized return.

Check. What conditions would make repeated returns identical?

Taking expectations of the return recursion

For this calculation, initialize the model at $X_t = x$ and follow π .

$$\begin{aligned} V^\pi(x) &= \mathbb{E}[G_t^\pi \mid X_t = x] \\ &= \mathbb{E}[R_t + \gamma G_{t+1}^\pi \mid X_t = x] \\ &= r^\pi(x) + \gamma \mathbb{E}[G_{t+1}^\pi \mid X_t = x] \\ &= \boxed{r^\pi(x) + \gamma \mathbb{E}[V^\pi(X_{t+1}) \mid X_t = x]}. \end{aligned}$$

For the last step, use the tower property: average a conditional expectation.

$$\mathbb{E}[G_{t+1}^\pi \mid X_t = x] = \mathbb{E}[\mathbb{E}[G_{t+1}^\pi \mid X_t, A_t, X_{t+1}] \mid X_t = x].$$

The stationary Markov model and policy give

$$\mathbb{E}[G_{t+1}^\pi \mid X_t, A_t, X_{t+1}] = V^\pi(X_{t+1}).$$

The last identity holds almost surely (with probability one). Conditional means agree; individual future returns can differ.

Expand the next-state expectation

At state x , sample $A \sim \pi(\cdot | x)$, then $X' \sim \mathcal{P}(\cdot | x, A)$.

$$\mathbb{E}[V^\pi(X')] = \int_{\mathcal{A}} \pi(da | x) \int_{\mathcal{X}} \mathcal{P}(dx' | x, a) V^\pi(x').$$

In a finite or countable model, the same expression is

$$\sum_{a, x'} \pi(a | x) \mathcal{P}(x' | x, a) V^\pi(x').$$

A compact form of the same two-stage average is $\int \mathcal{P}(dx' | x, a) \pi(da | x) V^\pi(x')$. Both forms average over the action and the transition.

Bellman expectation equation

Combining the return recursion with the next-state expectation gives

$$V^\pi(x) = r^\pi(x) + \gamma \int \mathcal{P}(dx' \mid x, a) \pi(da \mid x) V^\pi(x').$$

For finite spaces,

$$V^\pi(x) = \sum_a \pi(a \mid x) \left[r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V^\pi(x') \right].$$

The equation relates the value at the current state to expected values at possible next states, with the policy held fixed.

Policy-induced transitions and compact notation

Average the transition distributions over the policy's actions:

$$\mathcal{P}^\pi(x' | x) = \sum_a \pi(a | x) \mathcal{P}(x' | x, a) \quad (\text{finite states/actions}),$$

$$\mathcal{P}^\pi(dx' | x) = \int_{\mathcal{A}} \pi(da | x) \mathcal{P}(dx' | x, a) \quad (\text{kernel notation}).$$

The Bellman equation then becomes

$$V^\pi(x) = r^\pi(x) + \gamma \int_{\mathcal{X}} \mathcal{P}^\pi(dx' | x) V^\pi(x'),$$

$$\boxed{V^\pi = r^\pi + \gamma \mathcal{P}^\pi V^\pi.}$$

For a finite model, V^π, r^π are column vectors; matrix rows index the current state and columns index the next state. Multiplication by $\mathcal{P}^\pi$ averages next-state values under the policy.

Bellman equation for Q^π

$$\begin{aligned} Q^\pi(x, a) &= r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) V^\pi(x') \\ &= r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) \pi(da' \mid x') Q^\pi(x', a'). \end{aligned}$$

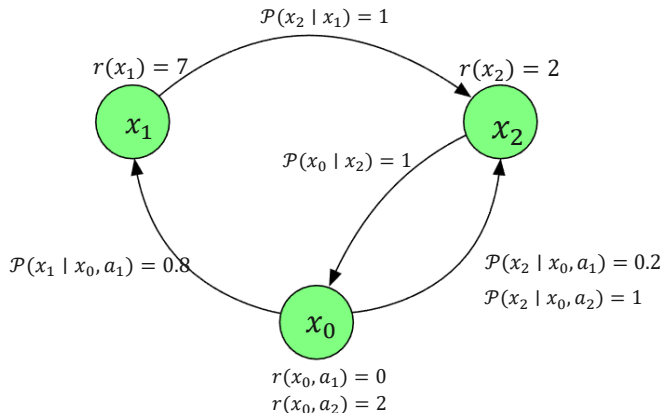
In compact form,

$$Q^\pi = r + \gamma \mathcal{P} V^\pi, \quad V^\pi(x) = \int \pi(da \mid x) Q^\pi(x, a).$$

Here $(\mathcal{P} V^\pi)(x, a) = \sum_{x'} \mathcal{P}(x' \mid x, a) V^\pi(x')$ is indexed by state–action pairs; $\mathcal{P}^\pi V^\pi$ is indexed only by states.

The current action a is fixed. The policy selects the next action a' after observing x' .

A three-state MDP and two policies



Two deterministic stationary policies

$$\pi_1(x_0) = a_1,$$

$$\pi_2(x_0) = a_2.$$

At x_1 and x_2 , both policies take the only available action.

Each policy repeats its rule on every visit to a state.

Continuing task: $\gamma = 0.9$.

Arrow labels give transition probabilities. Rewards belong to the current state/action.

First we evaluate π_1 together; then you will evaluate π_2 .

Worked example: evaluating π_1

$$\boxed{V^\pi = r^\pi + \gamma \mathcal{P}^\pi V^\pi} \quad V^\pi(x) = r^\pi(x) + \gamma \sum_{x'} \mathcal{P}^\pi(x' | x) V^\pi(x').$$

π_1 selects a_1 at x_0 , and the sole action at x_1, x_2 . Set $\pi = \pi_1$ and $\gamma = 0.9$:

$$\underbrace{\begin{bmatrix} V^{\pi_1}(x_0) \\ V^{\pi_1}(x_1) \\ V^{\pi_1}(x_2) \end{bmatrix}}_{V^{\pi_1}} = \underbrace{\begin{bmatrix} 0 \\ 7 \\ 2 \end{bmatrix}}_{r^{\pi_1}} + \underbrace{0.9}_{\gamma} \underbrace{\begin{bmatrix} 0 & 0.8 & 0.2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\mathcal{P}^{\pi_1}} \underbrace{\begin{bmatrix} V^{\pi_1}(x_0) \\ V^{\pi_1}(x_1) \\ V^{\pi_1}(x_2) \end{bmatrix}}_{V^{\pi_1}}.$$

Read one row at a time:

$$V^{\pi_1}(x_0) = 0 + 0.9[0.8V^{\pi_1}(x_1) + 0.2V^{\pi_1}(x_2)],$$

$$V^{\pi_1}(x_1) = 7 + 0.9V^{\pi_1}(x_2),$$

$$V^{\pi_1}(x_2) = 2 + 0.9V^{\pi_1}(x_0).$$

State order: (x_0, x_1, x_2) .

Matrix rows: current state.

Matrix columns: next state.

The first row $(0, 0.8, 0.2)$ weights the three next-state values.

r^{π_1} is known; V^{π_1} is unknown.

Worked example: solving for V^{π_1}

$$\boxed{V^{\pi} = r^{\pi} + \gamma \mathcal{P}^{\pi} V^{\pi}} \quad V^{\pi}(x) = r^{\pi}(x) + \gamma \sum_{x'} \mathcal{P}^{\pi}(x' | x) V^{\pi}(x').$$

Substitute the x_1 equation, then the x_2 equation.

$$\begin{aligned} V^{\pi_1}(x_0) &= 0.9[0.8(7 + 0.9V^{\pi_1}(x_2)) \\ &\quad + 0.2V^{\pi_1}(x_2)] \\ &= 5.04 + 0.828V^{\pi_1}(x_2) \\ &= 5.04 + 0.828(2 + 0.9V^{\pi_1}(x_0)) \\ &= 6.696 + 0.7452V^{\pi_1}(x_0). \end{aligned}$$

$$0.2548V^{\pi_1}(x_0) = 6.696 \implies V^{\pi_1}(x_0) \approx 26.28.$$

Substitute back:

$$\begin{aligned} V^{\pi_1}(x_2) &= 2 + 0.9V^{\pi_1}(x_0) \approx 25.65, \\ V^{\pi_1}(x_1) &= 7 + 0.9V^{\pi_1}(x_2) \approx 30.09. \end{aligned}$$

$$\boxed{V^{\pi_1} \approx (26.28, 30.09, 25.65)^{\top}}.$$

Round only at the end; $^{\top}$ denotes transpose.

Known quantities

$\pi_1(x_0) = a_1$, every visit.

At x_1, x_2 : the sole action.

$\gamma = 0.9$; continuing task.

$$\underbrace{\begin{bmatrix} 0 \\ 7 \\ 2 \end{bmatrix}}_{r^{\pi_1}} \quad \underbrace{\begin{bmatrix} 0 & 0.8 & 0.2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\mathcal{P}^{\pi_1}}$$

State order: (x_0, x_1, x_2) .

Rows: current state.

Columns: next state.

Class quiz: evaluate π_2

$$\boxed{V^\pi = r^\pi + \gamma \mathcal{P}^\pi V^\pi}$$

$$V^\pi(x) = r^\pi(x) + \gamma \sum_{x'} \mathcal{P}^\pi(x' | x) V^\pi(x').$$

Your turn: 4 minutes.

1. Substitute the given $r^{\pi_2}, \gamma, \mathcal{P}^{\pi_2}$.

Write one Bellman equation for each matrix row.

$$V^{\pi_2}(x_0) = \underline{\hspace{4cm}}$$

$$V^{\pi_2}(x_1) = \underline{\hspace{4cm}}$$

$$V^{\pi_2}(x_2) = \underline{\hspace{4cm}}$$

2. Solve for the three state values.

$$V^{\pi_2} = \begin{bmatrix} \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} \end{bmatrix} \begin{matrix} x_0 \\ x_1 \\ x_2 \end{matrix}$$

Known quantities

$\pi_2(x_0) = a_2$, every visit.

At x_1, x_2 : the sole action.

$\gamma = 0.9$; continuing task.

$$\underbrace{\begin{bmatrix} 2 \\ 7 \\ 2 \end{bmatrix}}_{r^{\pi_2}} \quad \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\mathcal{P}^{\pi_2}}$$

State order: (x_0, x_1, x_2) .

Rows: current state.

Columns: next state.

Class quiz: solution for π_2

$$\boxed{V^\pi = r^\pi + \gamma \mathcal{P}^\pi V^\pi} \quad V^\pi(x) = r^\pi(x) + \gamma \sum_{x'} \mathcal{P}^\pi(x' | x) V^\pi(x').$$

Substitute $r^{\pi_2}, \gamma, \mathcal{P}^{\pi_2}$, one row at a time:

$$V^{\pi_2}(x_0) = 2 + 0.9V^{\pi_2}(x_2),$$

$$V^{\pi_2}(x_1) = 7 + 0.9V^{\pi_2}(x_2),$$

$$V^{\pi_2}(x_2) = 2 + 0.9V^{\pi_2}(x_0).$$

Substitute the x_2 equation into the x_0 equation:

$$V^{\pi_2}(x_0) = 2 + 0.9(2 + 0.9V^{\pi_2}(x_0))$$

$$= 3.8 + 0.81V^{\pi_2}(x_0),$$

$$0.19V^{\pi_2}(x_0) = 3.8 \implies V^{\pi_2}(x_0) = 20.$$

$$V^{\pi_2}(x_2) = 2 + 0.9(20) = 20, \quad V^{\pi_2}(x_1) = 7 + 0.9(20) = 25.$$

$$\boxed{V^{\pi_2} = (20, 25, 20)^\top}$$

Known quantities

$\pi_2(x_0) = a_2$, every visit.

At x_1, x_2 : the sole action.

$\gamma = 0.9$; continuing task.

$$\underbrace{\begin{bmatrix} 2 \\ 7 \\ 2 \end{bmatrix}}_{r^{\pi_2}} \quad \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{\mathcal{P}^{\pi_2}}$$

State order: (x_0, x_1, x_2) .

Rows: current state.

Columns: next state.

One different action, then the same policy

At x_0 , take a_1 once and follow π_2 afterward:

$$\begin{aligned}Q^{\pi_2}(x_0, a_1) &= 0 + 0.9 [0.8V^{\pi_2}(x_1) + 0.2V^{\pi_2}(x_2)] \\&= 0.9[0.8(25) + 0.2(20)] = 21.6.\end{aligned}$$

Following π_2 from the first action instead gives

$$Q^{\pi_2}(x_0, a_2) = V^{\pi_2}(x_0) = 20.$$

Thus a smaller immediate reward can lead to a larger action value.

The value 21.6 describes one deviation followed by π_2 , including action a_2 on later visits to x_0 . Choosing a_1 on every visit instead follows π_1 , whose value at x_0 is approximately 26.28, not 21.6.

Optimality and Greedy Policies



The optimality equation and its three claims

$$V^*(x) = \max_{a \in \mathcal{A}} \left\{ r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) V^*(x') \right\}.$$

For finite discounted MDPs, three facts justify this characterization:

1. This equation has a unique bounded solution.
2. Its solution equals the best value over stationary policies.
3. A stationary policy attaining that value is also optimal among nonstationary and history-dependent policies.

Existence of a fixed point and optimality of that fixed point are different claims. The complete optimality verification is provided in the optional continuation.

Bellman optimality for action values

$$Q^*(x, a) = r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) \max_{a' \in \mathcal{A}} Q^*(x', a').$$

The first action is fixed; later decisions are optimal.

$$V^*(x) = \max_a Q^*(x, a).$$

Compare with policy evaluation:

$$\underbrace{\int \pi(da' \mid x') Q^\pi(x', a')}_{\text{follow the specified policy}} \longrightarrow \underbrace{\max_{a'} Q^*(x', a')}_{\text{choose the best next action}}.$$

The next-state transition is still averaged using $\mathcal{P}$.

Questions about the Bellman equations

We now have equations for

$$V^\pi, \quad Q^\pi, \quad V^*, \quad Q^*.$$

- What guarantees a unique solution to each equation?
- How can we compute a solution?
- How does a value function determine a policy?

The uniqueness conclusions have been stated; their justification comes next. We first answer the policy question with greedy action selection. Then Bellman operators and contraction give tools for the other two.

Recover an optimal policy from optimal values

$\max$ gives the largest value; argmax gives the set of actions that achieve it. The symbol $\in$ selects one action from that set:

$$\begin{aligned}\pi^*(x) &\in \operatorname{argmax}_{a \in \mathcal{A}} Q^*(x, a) \\ &= \operatorname{argmax}_{a \in \mathcal{A}} \left\{ r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) V^*(x') \right\}.\end{aligned}$$

With consistent tie-breaking, this defines a deterministic stationary policy.

To use V^* , evaluate a one-step transition. To use Q^* , compare the action values directly.

Optimal action now, optimal continuation later

$$\pi^*(x) \in \operatorname{argmax}_a \left\{ r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) V^*(x') \right\}.$$

1. Suppose future decisions use an optimal policy.
2. If the next state is x' , those decisions have value $V^*(x')$.
3. The next state is uncertain, so average its value using $\mathcal{P}$.
4. Add the current reward and select the maximizing action now.

The continuation value already includes the future sequence of decisions.

Greedy policies for candidate values

Let $\mathcal{B}(\Omega)$ denote bounded real-valued functions on Ω . In a finite MDP, these are tables of finite numbers. For arbitrary candidates $Q \in \mathcal{B}(\mathcal{X} \times \mathcal{A})$ and $V \in \mathcal{B}(\mathcal{X})$,

$$\pi_g(x; Q) \in \operatorname{argmax}_a Q(x, a),$$

$$\pi_g(x; V) \in \operatorname{argmax}_a \left\{ r(x, a) + \gamma \int \mathcal{P}(dx' \mid x, a) V(x') \right\}.$$

We also write $\pi_g(Q)$ or $\pi_g(V)$ for the entire policy. For optimal values and consistent tie-breaking,

$$\pi_g(V^*) = \pi_g(Q^*) = \pi^*.$$

Why a greedy decision can be optimal

- Greedy selection with Q compares the numbers $Q(x, a)$.
- Greedy selection with V examines one immediate transition.
- Neither selection procedure explicitly enumerates all future trajectories.

When the candidate function is Q^* or V^* , its values already encode optimal continuation. That is why a local comparison suffices.

Caution. A greedy policy for an arbitrary candidate value is not necessarily optimal. Q^π assumes future actions follow π . Changing the policy at every state changes that continuation; its new value must be evaluated accordingly.